

4.2 Database Components (Tables, Records, Fields)

ٹیبل قطاروں اور کالموں میں ڈیٹا رکھتا ہے۔

CS XI

Chapter 4: Data and Analysis

Lecture 03 of 15

4.2 Database Components (Tables, Records, Fields)

LEARN IT

The basic structure is a **table** (also called a relation): rows and columns, like a spreadsheet, holding facts about one kind of thing (Students, Teachers, Books).

A **record** (row, tuple) is one complete instance — everything about Ali. A **field** (column, attribute) is one kind of fact — the whole Name column, or the whole Grade column.

Roll No	Name	Course	Grade
101	Ali	CS	A
102	Sara	IT	B+
103	Ahmed	CS	A

The row 101 | Ali | CS | A is one **record**. The Name column is one **field**.

REMEMBER IT

Memory: **row = record = one person. column = field = one question on the admission form.**

COMMON MISTAKE

Do not mix two entities in one table (students and library books together). Each entity gets its own table; keys will link them later.

اردو میں سمجھیں

ٹیبل قطاروں اور کالموں میں ڈیٹا رکھتا ہے۔ ایک قطار ایک ریکارڈ (مثلاً علی کی پوری معلومات) ہے، ایک کالم ایک فیلڈ (مثلاً صرف نام) ہے۔

CHECK YOURSELF

1. How many records and how many fields are in the table above?
2. Is "CS" a record or a field value?

Answers: 1. 3 records, 4 fields. 2. A value stored in the Course field.

Next lecture: 4.3.1 Keys and Integrity Constraints